

Online 1: Section C (C1 + C2)

Instructions

- Follow the function signatures provided in the starter code.
- Do not change variable names used in the training loop.
- You may use NumPy vectorized operations.

C1: Core Functions

1. **Mean Imputation** (column-wise): Replace missing values in X with the mean of each column.
2. **Leaky ReLU**: Implement the activation $f(x) = x$ for $x > 0$ and $f(x) = \alpha x$ otherwise.
3. **Leaky ReLU Gradient**: Implement $\frac{\partial f}{\partial x}$ for the activation in part 2.
4. **Huber Loss**: Implement

$$L_{\delta}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & |y - \hat{y}| \leq \delta \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2, & \text{otherwise} \end{cases}$$

5. **Huber Loss Gradient**: Implement $\frac{\partial L_{\delta}}{\partial \hat{y}}$.

C2: Training and Inference

1. Use mini-batch gradient descent to train the linear model with Leaky ReLU output.
2. Report training loss and RMSE at the end of each epoch.
3. Implement inference on `test_data.csv` and report the test RMSE.